CENTRALE NANTES

ENGINEERING PROGRAMME

2024-2025

Year 2 / Year 3

Specialisation option

Data Analysis and Applications in Signal and Image Processing

OD DATASIM

PROGRAMME SUPERVISOR
Said MOUSSAOUI

Autumn Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 73	12	Core course	ARSIG ATRIM CSOPT MSTAT	Signal representation and analysis Image processing and analysis Scientific computing and numerical optimization Statistical data modelling and analysis
UE 74	13	Core course	APSTA FEBIO GRASP IMINV PRTSI1	Machine learning theory and practice Filtering and biomedical signal processing Graph theory and time series analysis Imaging and Inverse Methods Project in signal and image processing

Spring Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 83	14	Core course	AIMED AMULT APPLI AUDIO PRTSI 2	Advanced machine learning and medical imaging Multi-sensor data analysis R&D applications Audio content analysis and Information Retrieval Project in signal and image processing

Signal representation and analysis [ARSIG]

LEAD PROFESSOR(S): Sébastien BOURGUIGNON

Requirements

Mathematical fouondations: linear algebra, integration, series, numerical optimization. Fourier series, Fourier transform.

Objectives

Extracting relevant information from a data set is a key step for efficient data processing. For example, the use of specific analysis methods is crucial for event detection in a noisy environment, extraction of characteristic features for statistical learning and classification, noise reduction or data compression.
This course introduces a set of mathematical (analytical and numerical) tools for representing a signal, an image or, more generally, a data set, in order to extract its meaningful and useful information. Different analysis tools are presented, both in their mathematical and informational foundations and in their practical numerical implementation, through application examples taken from real data processing problems.
The first part of the course deals with the analysis of stationary signals, through Fourier analysis and high-resolution spectral analysis. Time-frequency and time-scale representations are studied for the analysis of non-stationary data. Finally, some elements concerning the more recent sparse representation framework are introduced, which generalise the former representations to more complex issues, in particular adapted to modern problems concerning big data processing.

Course contents

1. Spectral analysis for stationary signals: Fourier analysis, high-resolution methods. Lab work: detection of multiple oscillating components in noise; application to exoplanet detection from time series.

2. Time-frequency analysis: linear (short-term Fourier transform) and quadratic (spectrogram, Wigner-Ville) representations. Numerical tutored work: comparison of different time-frequency representations; time-frequency representations of piano instruments.

3. Non-stationary signals: time-scale representations and wavelet transforms. Lab work: Discrete Wavelet Transform and multiscale analysis; application to signal denoising and image compression.

4. Toward more general models: sparse representations. Lab work: denoising through projection into orthogonal bases and into redundant spaces ; denoising of piecewise constant signals.

Course material

A.V. Oppenheim and R.W. Schafer. Discrete-time signal processing, Prentice Hall, 2010.
S. Marcos. Les méthodes à haute résolution : Traitement d'antenne et analyse spectrale, Hermès, 1998.

L. Cohen, Time-Frequency analysis, Prentice-Hall, 1995.
N. Martin et C. Doncarli, Décision dans le plan temps-fréquence, Hermes, 2004.

S. Mallat, A Wavelet Tour of Signal Processing: The Sparse Way, Academic Press, 2008.

M. Elad, Sparse and Redundant Representations, Springer, 2010.

Assessment

Collective assessment:	EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	8 hrs	12 hrs	10 hrs	0 hrs	2 hrs

ENGINEERING - OD DATASIM

Year 2 / Year 3 - Autumn Semester - Course Unit 73 / 93

Image processing and analysis [ATRIM]

LEAD PROFESSOR(S): Diana MATEUS LAMUS

Requirements

Objectives

This course is an introduction to the essential techniques for digital image processing. It covers how images are acquired, formed, stored, processed and used. We will review different types of image processing methods, including techniques for image transformation, enhancement, filtering, denoising, and some segmentation bricks. The course is composed of a series of lectures accompanied by significant hands-on experience to put into practice and analyse the taught techniques.

Course contents

Content:
- Introduction to digital images
- Quantitative image quality metrics
- Morphological operators
- Spatial filtering
- Spectral filtering
- Geometric transformations
- Segmentation methods

Course material

[1] Digital Image Processing, 4th Ed. Gonzalez and Woods © 2018, ISBN: 9780133356724
[2] Computer Vision: Algorithms and Applications (Texts in Computer Science). 2011th Edition. by Richard Szeliski
[3] D. Forsyth, J. Ponce, Computer vision: a modern approach, Prentice Hall 2003.

Assessment

Collective assessment: EVC 1 (coefficient 0.5)

Individual assessment: EVI 1 (coefficient 0.5)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	8 hrs	12 hrs	10 hrs	0 hrs	2 hrs

Scientific computing and numerical optimization [CSOPT]

LEAD PROFESSOR(S): Said MOUSSAOUI

Requirements

Objectives

This course focuses on scientific computing and numerical optimization methods that are used in signal and image processing. It firstly provides a theoretical description of the methods and then addresses some examples such as non-linear curve fitting and signal/image denoising.

Course contents

1. Scientific computing
- Matlab
- Python

2. Unconstrained optimization
- basic concepts: differential calculus, mathematical properties, optimality conditions
- iterative methods (first and second order methods, descent direction, line search, trust region)
- applications (usual functions, parametric curve fitting, signal denoising)

3. Constrained optimization
- exterior penalty methods
- interior-point methods
- applications to image restoration

4. Global optimization
- interval methods
- evolutionary methods
- Monte Carlo methods

Course material

[1] J. Nocedal and S. J. Wright. Numerical Optimization. Springer series in operations research, Springer, 1999
[2] S. Boyd and L. Vendenberghe. Convex Optimization. Cambridge University Press, 2004
[3] P. Venkataraman. Applied Optimization with Matlab Programming, John Wiley and Sons, 2001

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	8 hrs	12 hrs	10 hrs	0 hrs	2 hrs

ENGINEERING - OD DATASIM

Year 2 / Year 3 - Autumn Semester - Course Unit 73 / 93

Statistical data modelling and analysis [MSTAT]

LEAD PROFESSOR(S): Eric LE CARPENTIER

Requirements

Objectives

This course addresses the characterization and the processing of random signals by means of statistical tools. It provides the theoretical foundations used in practical problems to estimate a quantity of interest and to retrieve sought information.
Applications concern: biomedical signal and image processing (diagnosis, tools to assist the disabled), music signal processing (recording, restoration, coding), positioning systems, etc.
At the end of the course the students will be able to:

- Provide a statistical description of a random process
- Solve a statistical estimation problem in a practical situation
- Derive a numerical algorithm to calculate and to characterize the solution

Course contents

- Probability theory: random vectors, density, mean, variance.
- Time analysis, frequency analysis: random signals, autocorrelation, power spectral density.
- Classical estimation, Bayesian estimation: maximum likelihood (ML) estimation, minimum mean square error (MMSE) estimator, maximum a posteriori (MAP) estimator, linear minimum mean square error (LMMSE).
- Markov chains, Markov processes.
- Statistical filtering: Bayes, Kalman, particles.

Course material

[1] Probability, Random Variables and Stochastic Processes. A. Papoulis, S.U. Pillai. Mc Graw Hill.
[2] Fundamentals of Statistical Signal Processing, Vol.1: Estimation theory, S. Kay, Prentice Hall.

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	10 hrs	10 hrs	0 hrs	2 hrs

ENGINEERING - OD DATASIM

Year 2 / Year 3 - Autumn Semester - Course Unit 74 / 94

Machine learning theory and practice [APSTA]

LEAD PROFESSOR(S): Diana MATEUS LAMUS

Requirements

Objectives

The aim of this course is to provide students with the key notions of machine learning, essential today in dealing with the ubiquitous collection of increasing amounts of data. The course will introduce the different types of machine learning and their applications in the context of signals and images. We will review the most influential methods for unsupervised and supervised learning. The sessions will alternate between lectures, practical exercises, and mini-projects in Python. Although the techniques will be presented from a broad and general perspective, the applications will focus on signal and images analysis, in particular from biomedical fields.

Course contents

- Introduction to machine learning
- Supervised and unsupervised methods
- Data representation, variable selection, and dimensionality reduction
- Evaluation measurements
- Probabilistic and linear classification methods
- Support Vector Machines (SVM)
- Decision trees and random forests
- Neural networks
- Introduction to deep learning
- Semantic Segmentation

Course material

[1] Bishop C. : Pattern Recognition and Machine Learning. Springer, 2006.
[2] Kevin P. Murphy, Probabilistic Machine Learning: An Introduction, 2022

Assessment

Collective assessment: EVC 1 (coefficient 0.5)

Individual assessment: EVI 1 (coefficient 0.5)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	10 hrs	10 hrs	0 hrs	2 hrs

Filtering and biomedical signal processing [FEBIO]

LEAD PROFESSOR(S): Said MOUSSAOUI

Requirements

Objectives

- learn how to synthesize a linear filter respecting the predefined requirements
- make some application of filtering to real data processing
- have a background view of biomedical signal acquisition and processing

Course contents

1/ Linear filtering
- characterization of filters (specifications on frequency response)
- linear filter synthesis methods
- IIR and FIR filter synthesis

2/ Biomedial signal porcessing
- Electrocardiography: acquisition principles and processing methods
- Electromyography: acquisition principles and processing methods
- Electroencephalography: acquisition principles and processing methods

Course material

[1] M. Najim, Digital Filters Design for Signal and Image Processing, 2013, Wiley
[2] A. Zverev, Handbook of Filter Synthesis, John Wiley and Sons, 1967
[3] Naik Ganesh R., Santos Wellington Pinheiro Dos. Biomedical Signal Processing. A Modern Approach. Lavoisier, 2023

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	12 hrs	8 hrs	0 hrs	2 hrs

Graph theory and time series analysis [GRASP]

LEAD PROFESSOR(S): Mira RIZKALLAH

Requirements

- Basics on statistics and probability
- Mathematical notions : complex numbers, integrals, linear regression, machine learning and deep learning

Objectives

At the end of this course, students will know:
▪ Estimate the parameters of time series models (AR, MA, ARMA, ARIMA, etc.)
▪ Prediction of future values and corresponding uncertainties from estimated models
▪ Know the basic principles of signal processing on graph: Fourier transform on graph, filtering, statistical and deep learning on graphs, convolutional networks on graphs.
▪ Apply signal processing and learning methods to a problem and data real (neuroscience, brain connectivity)

Course contents

The first part of the course focuses on experimental modeling of time series. It provides a detailed description of the time series forecasting and modeling chain, from data analysis to model validation. The second part covers the principles of graph signal processing and graph convolutional neural networks.

1. Time series analysis
- Basics of time series: important characteristics of a time series graph, auto-regressive (AR) model, autocorrelation function (ACF), moving average (MA) models, study of residuals.
- Moving average (MA) models, partial autocorrelation, useful notations (lag operators..)
- Identification and estimation of ARMA, ARIMA models; use of ARIMA models to forecast future values.
- Seasonal ARIMA models and exponential smoothing

2. Signal processing and learning on graphs
- Basic notions of graph-based modeling, signal on graphs, applications, the interest of representation, how to define a graph?
- Graph fourier transform, convolution and filtering of signals on graphs
- Neural networks on graphs, theory and applications in different fields

3. Practical applications
- Prediction of climate change with ARIMA, SARIMA models. In-depth analysis of models and data transformations.
- Node and Graph Classification. Applications on real data: social networks and neuroscience

Course material

▪ H. Kwakernaak and R. Sivan, Modern signals and systems, Prentice Hall, Englewood Cliffs, 1991
▪ C. Chatfield and H. Xing, The Analysis of Time Series An Introduction with R, by Chapman and Hall/CRC, 2019
▪ Antonio Ortega, Introduction to graph signal processing , Cambridge University Press.

Assessment

Collective assessment:	EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	10 hrs	10 hrs	0 hrs	2 hrs

Imaging and Inverse Methods [IMINV]

LEAD PROFESSOR(S): Sébastien BOURGUIGNON

Requirements

Basics of statistical analysis and probability theory. Linear algebra: matrix manipulation, matrix properties. Numerical optimization.

Objectives

Inverse methods are used when information about an object under study is acquired indirectly, by measuring the effects of a physical phenomenon whose causes are sought. This measurement principle can be found in many applications: optics, acoustics, medical imaging, oceanography, astronomy, non-destructive testing of materials, etc. This is the case for most imaging devices, where the aim is to map physical properties inside an object under study from measurements located at its periphery (e.g., X-ray tomography, or CT scan). This is also the case for methods aiming to restore a signal, an image, a data set from a version that was downgraded, distorted and filtered by the acquisition device (e.g., PSF of a microscope or of a telescope).

This course aims to introduce students to basic and more advanced tools to address an inversion problem, from the definition of the direct problem (physical modelling) and general information theory elements (incomplete data, ill-posed problem, statistical inference), to the resolution principle essentially based on regularization, up to the efficient numerical computation of the solution by means of specific algorithms.

Course contents

1. General points: ill-posed problems, regularization, a priori information

2. Deconvolution: standard (linear) methods, quadratic regularization

3. Non-linear methods, spike train deconvolution, sparsity

4. Image restoration and tomography

Course material

A. Tarantola, Inverse Problem Theory and Model Parameter Estimation, SIAM, 2005.
J. Idier (Ed.), Bayesian Approach to Inverse Problems, ISTE Ltd and John Wiley & Sons Inc, 2008.
M. Bertero, P. Boccacci, Introduction to Inverse Problems in Imaging, CRC Press, 1998
P.C. Hansen, Discrete Inverse Problems: Insight and Algorithms, SIAM, 2010
J. M. Mendel, Optimal Seismic Deconvolution, Academic Press, 1983.
A. C. Kak et M. Slaney, Principles of Computerized Tomographic Imaging, IEEE Press, 1988.

Assessment

Collective assessment:	EVC 1 (coefficient 0.3)
Individual assessment:	EVI 1 (coefficient 0.7)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	8 hrs	12 hrs	0 hrs	2 hrs

Project in signal and image processing [PRTSI1]

LEAD PROFESSOR(S): Mira RIZKALLAH

Requirements

Objectives

The aim of this course is to undertake a project related to the processing of real data: audio signals, biomedical images or signals, hyperspectral images etc. This project begins in the autumn semester and is pursued in the spring semester.

Course contents

The projects concern the processing of real data such as:

1. Spectral data and hyperspectral images: deconvolution, decomposition and separation
2. Physiological signals: EEG, EMG for BCI and Prosthesis control
3. Audio signal: segmentation, speech recognition, content analysis
4. Medical image analysis: image segmentation, pattern recognition, etc

Course material

[1] Fundamentals of statistical signal processing - Vol I. Estimation theory. S. KAY. Prentice Hall, 1993.
System identification, theory for the user. L. LJUNG Prentice Hall, Englewood Cliffs, New Jersey, 1987 (1st ed.) - 1999 (2nd ed.).
[2] Approche bayésienne pour les problèmes inverses. J. IDIER. Traité IC2, Série traitement du signal et de l'image, Hermès, 2001.
[3] Pattern Classification. R.O. DUDA, P.E. HART, D.G.STORK, Willey 2001.
Analyse d'images, filtrage et segmentation. Sous la direction de J.P. COQUEREZ et S. PHILIPP, Masson 1995

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	1	0 hrs	0 hrs	0 hrs	32 hrs	0 hrs

Year 2 / Year 3 - Spring Semester - Course Unit 103 / 83

Advanced machine learning and medical imaging [AIMED]

LEAD PROFESSOR(S): Diana MATEUS LAMUS

Requirements

Objectives

This course will address advanced aspects of acquiring, reconstructing, and processing biomedical signals and images. The module aims to put in practice concepts acquired in previous periods (optimization approaches, data analysis algorithms, machine learning methods, etc.) in various practical application contexts. Several speakers from the industrial and research sectors will present real-world challenges they are working on and will propose mini-projects to be solved during practical sessions. These thematic sessions will be accompanied by additional lectures and practical sessions, providing alternative solutions

Course contents

Introduction to various medical imaging modalities such as ultrasound, computed tomography (CT), positron emission tomography (PET), or magnetic resonance imaging (MRI).

Practical implementation of algorithms for image formation/reconstruction, noise reduction, classification, segmentation, registration, and anomaly detection.

Course material

Medical Imaging Signals and Systems. Prince Links. Pearson Editions. 2015

Assessment

Collective assessment: EVC 1 (coefficient 1)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	14 hrs	8 hrs	0 hrs	0 hrs

Multi-sensor data analysis [AMULT]

LEAD PROFESSOR(S): Said MOUSSAOUI

Requirements

Objectives

This course focuses on techniques for analysing multi-sensor data using so-called source separation methods. These recent methods - principal component analysis, independent component analysis and non-negative matrix factorization - exploit sensor redundancy to analyse data content by exploiting linear mixture models and adopting statistical assumptions about the sources.

The second part of the course addresses applications for signal and image processing in order to show how to adapt the analysis methods to the constraints of the applications.

Course contents

1. Source separation methods
a) Algebraic methods
b) Iterative methods
c) Statistical approaches (maximum likelihood and Bayesian estimation)

2. Applications for signal analysis
a) Spectrometry
b) Biomedical signals

3. Applications for hyperspectral imaging
a) Concepts of remote sensing
b) Geometrical approaches
c) Hyperspectral image analysis by source separation

Course material

[1] P. Comon and C. Jutten, Séparation de sources 1: concepts de base et analyse en composantes indépendantes, Traité IC2, série Signal et image, 03-2007
[2] P. Comon and C. Jutten, Séparation de sources 2: au-delà de l'aveugle et applications, Traité IC2, série Signal et image, 03-2007
[3] A. Hyvarinen, J. Karhunan, and E. Oja, Independent Component Analysis. Johns Willey & Sons., 2001.

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	10 hrs	10 hrs	0 hrs	2 hrs

Year 2 / Year 3 - Spring Semester - Course Unit 103 / 83

R&D applications [APPLI]

LEAD PROFESSOR(S): Sébastien BOURGUIGNON

Requirements

Objectives

This course is composed of interventions given by R&D engineers and researchers working on data analysis and processing problems in various fields of science and engineering. It aims at presenting how different kinds of information processing problems (signal, image, data analysis) are tackled in several industrial or academic research activities, using modern methodologies, in particular using the various courses and tools seen in the first two periods of the year of the DATASIM track.

Course contents

Each course takes place over one day. The morning is dedicated to a presentation of the problem encountered, its methodological obstacles and the solutions that are proposed. The afternoon is dedicated to the practical implementation of the methods developed in the first part, on real data. Provisional program (subject to changes):

ONERA: Co-design approaches for the joint acquisition and processing of data, and applications in optical imaging

RENAULT: Introduction to Data Fusion for Driving Assistance and Autonomous Vehicles

DB-SAS: Ultrasound imaging for non-destructive testing

Université Gustave Eiffel : Image Processing using LBM: Anisotropic Diffusion and Segmentation

Lagrange Laboratory (Observatory of the Côte d'Azur): Introduction to detection and some applications in Astrophysics

CNES: Space imaging: from principles to applications

Course material

Ku, Jason, et al. "Joint 3d proposal generation and object detection from view aggregation." 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2018.
S. Steyer, G. Tanzmeister and D. Wollherr, "Object tracking based on evidential dynamic occupancy grids in urban environments," 2017 IEEE Intelligent Vehicles Symposium (IV), Los Angeles, CA, 2017, pp. 1064-1070, doi: 10.1109/IVS.2017.7995855.

"Fundamentals of Statistical Signal Processing, Volume II: Detection Theory", S. Kay, Prentice Hall, 1993, ISBN-10: 013504135X

Valorge C. and al., (2012) Satellite Imagery. From acquisition principles to processing of optical images for observing the Earth, Cépadues Edition
Baghdadi N. and al., (2016). Optical Remote Sensing of Land Surface: Techniques and Methods, ISTE Press Ltd - Elsevier Inc

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	16 hrs	16 hrs	0 hrs	0 hrs	0 hrs

Audio content analysis and Information Retrieval [AUDIO]

LEAD PROFESSOR(S): Jean-François PETIOT

Requirements

Tools for audio signal processing

Objectives

This course aims to train students in:
- the principles of audio numerical representation (acoustic, encoding, processing)
- the principles of psycho acoustics (perception, cognition)
- modelling and learning of high level attributes (musical information retrieval)

Course contents

Basics of Acoustics
Representation of sounds
- spectrogramme, spectral analysis
Sound Processing and data analysis
- PCA, MDS
Perception and psychoacoustics
Auditory Scene Analysis
Source separation

Course material

Acoustique, Informatique et Musique. Outils scientifiques pour la musique. B. D'Andréa-Novel, B. Fabre, P. Jouvelot. Mines Paristech, 2012
Klapuri, A. and Davy, M. (Editors), Signal Processing Methods for Music Transcription, Springer-Verlag, New York, 2006
Auditory Scene Analysis, The Perceptual Organization of Sound, By Albert S. Bregman, MIT Press

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	12 hrs	8 hrs	0 hrs	2 hrs

Year 2 / Year 3 - Spring Semester - Course Unit 103 / 83

Project in signal and image processing [PRTSI 2]

LEAD PROFESSOR(S): Mira RIZKALLAH

Requirements

Objectives

The aim of this course is to undertake a project related to the processing of real data: audio signals, biomedical images or signals, hyperspectral images. This project begins in the autumn semester and is pursued in the spring semester.

Course contents

The projects involve the processing of real data such as:

1. Spectral data and hyperspectral images: deconvolution, decomposition and separation
2. Physiological signals: EEG, EMG for BCI and Prosthesis control
3. Audio signals: segmentation, speech recognition, content analysis
4. Medical image analysis: image segmentation, pattern recognition, etc

Course material

[1 Fundamentals of statistical signal processing - Vol I. Estimation theory. S. KAY. Prentice Hall, 1993.
System identification, theory for the user. L. LJUNG Prentice Hall, Englewood Cliffs, New Jersey, 1987 (1st ed.) - 1999 (2nd ed.).
[2] Approche bayésienne pour les problèmes inverses. J. IDIER. Traité IC2, Série traitement du signal et de l'image, Hermès, 2001.
[3] Pattern Classification. R.O. DUDA, P.E. HART, D.G.STORK, Willey 2001.
Analyse d'images, filtrage et segmentation. Sous la direction de J.P. COQUEREZ et S. PHILIPP, Masson 1995

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	2	0 hrs	0 hrs	0 hrs	48 hrs	0 hrs

ENGINEERING PROGRAMME

2024-2025

Year 2 / Year 3

Specialisation option

Computer Science for Information Systems

OD INFOSI

PROGRAMME SUPERVISOR
Jean-Yves MARTIN

ENGINEERING - OD INFOSI

Autumn Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 73	12	Core course	BDONN GELOG MADIS OBJET	DataBases Software Engineering Discrete Mathematics Object Oriented Programming
UE 74	13	Core course	ADATA MEDEV_INFOSI PAPPL PRWEB SECUR	Data Analysis Industrial Software Development Project: Software Development Project Web Programming Systems and Data security

Spring Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 83	14	Core course	DEVMO PGROU SYRES SYSIN TLANG	UI-UX Design and mobile Dev Group Project Systems and Networks Information Systems Language Theory

ENGINEERING - OD INFOSI

Year 2 / Year 3 - Autumn Semester - Course Unit 73 / 93

DataBases [BDONN]

LEAD PROFESSOR(S): Jean-Yves MARTIN

Requirements

Objectives

The objective of this course is to understand the functioning of databases, from both theoretical and practical perspectives. Starting from relational algebra, we study the conceptual modeling of a more or less well defined problem and its transformation into a relational model and its operations through administrative tools or software. The focus lies particularly on the treatment of ill-posed problems, or the exploitation of poorly designed databases in order to prepare engineers for real situations.

Course contents

This course includes lectures, exercices and practical work.

Lectures will follow the following programme:
- Introduction to Databases
- Relationnal Databases
 + Functional Modeling, Relational Modeling, Physical Modeling
 + Relational Algebra
 + Introduction to Normal Forms
 + Introduction to SQL
 + Programming databases with java, python and PL/SQL
- Notions of BI
- Introduction to noSQL and Big Data
 + Introduction to mongoDB
 + Introduction to Cassandra, CHEBOTKO Diagramme

Practical work consists in a projet that requires database modelling and creation, SQL requests, programming in java, triggers implementation, use of MongoDB database through python and java programs.

Course material

Assessment

Collective assessment:	EVC 1 (coefficient 0.5)
Individual assessment:	EVI 1 (coefficient 0.5)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	10 hrs	10 hrs	0 hrs	2 hrs

ENGINEERING - OD INFOSI

Year 2 / Year 3 - Autumn Semester - Course Unit 73 / 93

Software Engineering [GELOG]

LEAD PROFESSOR(S): Myriam SERVIÈRES

Requirements

Objectives

To acquire the fundamentals in Software Engineering and Project Management.

Course contents

The course is structured around several major themes:
- Software development cycles (specifications, life cycle, planning, quality, specifications, production, acceptance),
- Analysis, specification and design models with a particular emphasis on UML,
- Fundamentals of IT project management,
- Introduction to Agile development (Scrum) and DevOps.

Upon completion of the course, students are expected to be able to design and model software and write specifications.

Course material

Modélisation objet avec UML, Pierre-Alain Muller, Best of Eyrolles, 2005.
UML2 et les design patterns, Craig Larman, Pearson Education, 2005.
Software Engineering 8, Ian Sommerville, Addison Wesley, 2007.
Le génie logiciel et ses applications, Ian Sommerville, InterEdition, 1988.
Processus d'ingénieries du logiciel, méthodes et qualité, Claude Pinet, Pearson Education, 2002.
UML2, Benoit Charroux, Aomar Osmani, Yann Thierry-Mieg, Pearson Education, 2005.

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	22 hrs	8 hrs	0 hrs	0 hrs	2 hrs

Discrete Mathematics [MADIS]

LEAD PROFESSOR(S): Jean-Sebastien LE BRIZAUT

Requirements

Binary calculation.
Matrix calculation.
Basics of probability.

Objectives

The objective of this course is to provide a number of mathematical tools used in solving computer problems.
- To show the diversity of tools needed in information coding
- To introduce some elements of theoretical computer science underlying the other courses of the computer science specialisation.

Course contents

- Introduction to information coding
    - Theoretical foundations of coding theory and its history,
    - Application examples, associated algorithms.
    - Error correcting codes,
    - Cryptography.
- Graphs
    - Introduction to graph theory (shortest paths, minimum cover, flow, layout).

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	20 hrs	10 hrs	0 hrs	0 hrs	2 hrs

Object Oriented Programming [OBJET]

LEAD PROFESSOR(S): Jean-Marie NORMAND

Requirements

Objectives

The objective of this course is for students to be able to program in the Java object-oriented language. It will introduce the main concepts of object-oriented programming (encapsulation, inheritance, polymorphism) to model them using UML (Unified Modelling Language), and put them into practice with Java language.

Then, the course will focus on the major classes of data structures and algorithms based on the implementation in Java.

Finally, some specific mechanisms will be covered such as interfaces, abstraction, generics, exceptions and introspection.

The course consists of lectures as well as numerous practical sessions. During the lab work, students realize a project that evolves to integrate all the notions discussed in the classes, the project runs throughout the whole course.

Course contents

Introduction to Java
Object-oriented concepts
Data structures and how to use them in Java
Abstract classes and methods, Interfaces
Generics and Exceptions
Reflexivity/Introspection

Addendum:
Packages
Threads in Java
Graphical User Interface in Swing

Course material

Assessment

Collective assessment: EVC 1 (coefficient 0.5)

Individual assessment: EVI 1 (coefficient 0.5)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	10 hrs	0 hrs	20 hrs	0 hrs	2 hrs

Data Analysis [ADATA]

LEAD PROFESSOR(S): Mathieu RIBATET

Requirements

Objectives

Learn and implement classical data analysis methods

Course contents

1- Introduction to modeling and basic concepts, data visualization
2- Unsupervised classification
3 - Principal component analysis
4- Linear regression

Course material

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	14 hrs	18 hrs	0 hrs	0 hrs	0 hrs

Industrial Software Development [MEDEV_INFOSI]

LEAD PROFESSOR(S): Myriam SERVIÈRES

Requirements

Objectives

This course aims to provide methods and tools for developing industrial-quality software. This includes unit and integration tests, version management, code metrics, continuous integration services, and design patterns. It will also be an opportunity to extend the students' technical knowledge.

Lab work will use java language. All notions covered in lectures will be applied practically in lab sessions.

Course contents

- Group work in computer science
- Version management
- Software tests
- Unit testing
- Advanced build tools and continuous integration
- Code metrics

Course material

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	6 hrs	10 hrs	16 hrs	0 hrs	0 hrs

Project: Software Development Project [PAPPL]

LEAD PROFESSOR(S): Myriam SERVIÈRES

Requirements

Objectives

The aim of this project is to build an application, using the concepts covered during the computer science lectures.

Course contents

The project undertaken in pairs. Emphasis is placed on project management, the quality of the deliverable, documentation of the source code and the results.

Projects change every year. They can include web development, specific software development, etc.

Course material

Assessment

Collective assessment: EVC 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	1	0 hrs	0 hrs	0 hrs	32 hrs	0 hrs

Web Programming [PRWEB]

LEAD PROFESSOR(S): Jean-Marie NORMAND

Requirements

Objectives

The objective of this course is to provide students with the fundamentals of web programming.

It starts with HTML-CSS and JavaScript before witching to higher level frameworks.

Students will thus have to program a small but functional Web application using:
- Spring (Java)
- ReactJS/NodeJS

This course includes a number of practical sessions devoted to working with frameworks and building Web applications.

Course contents

This course includes lectures and lab work.

Lectures:
- Introducing HTML, CSS and Javascript
- PHP
- J2EE
- Web servers
- Notions of Web Programming Frameworks

Lab work:
- HTML, Javascript, AJAX
- SPRING
- ReactJS/NodeJS

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	4 hrs	2 hrs	24 hrs	0 hrs	2 hrs

Systems and Data security [SECUR]

LEAD PROFESSOR(S): Jean-Yves MARTIN

Requirements

Objectives

The objective of this course is to give students an understanding of computer security, cybersecurity, and copyright. For this, each of these aspects is covered by a professional in the field, who addresses the basic concepts, the tools used in practice, and illustrates his or her point with examples.

Course contents

- Introduction to internet security, main attacks, encryption, and main mechanisms. Introduction to LDAP authentication.
- Security from an administrative perspective. Security planning, RSSI, Using security audit.
- Security from a company audit perspective.
- Security from a technical point of view. Main attacks: how we can protect computers and software.
- Security from a legal point of view. CNIL, software licenses. RGPD.
- Personal data protection.

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	20 hrs	10 hrs	0 hrs	0 hrs	2 hrs

UI-UX Design and mobile Dev [DEVMO]

LEAD PROFESSOR(S): Vincent TOURRE

Requirements

Objectives

UI-UX : Learn the principles of creating a User Interface (UI) taking into account user experience (UX).
Mobile development: Learn the logic behind the creation of a mobile application for the Android platform.

Course contents

User Interfaces:
- History of interfaces
- Interaction styles
- User experience
- Visual perception
- Ergonomic criteria

Mobile development:
- Production process of an Android application
- Programming activities in JAVA

Interface/mobile development project to practise the concepts (two students).

Course material

Assessment

Collective assessment: EVC 1 (coefficient 0.4)

Individual assessment: EVI 1 (coefficient 0.6)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	12 hrs	10 hrs	8 hrs	0 hrs	2 hrs

Group Project [PGROU]

LEAD PROFESSOR(S): Jean-Yves MARTIN

Requirements

Objectives

The aim of this project is to have students work in groups in order to address design issues, code sharing, project planning, development.

Course contents

This course is a project undertaken in groups of 5 to 7 students.

Emphasis is placed on project reporting, project management, code sharing, the quality of the deliverable, documentation of the source code and the results.

Projects change every year. They can include web development, specific software development, etc.

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	2	0 hrs	0 hrs	0 hrs	48 hrs	0 hrs

ENGINEERING - OD INFOSI

Year 2 / Year 3 - Spring Semester - Course Unit 103 / 83

Systems and Networks [SYRES]

LEAD PROFESSOR(S): Jean-Yves MARTIN

Requirements

Objectives

This course aims to provide the fundamentals of systems and networks. The first part of the course defines what is an operating system, the services to be expected and principal components. The second part of the course presents the problems of data-processing networks (general concepts, overview, challenges, customer-server, groupware, security).

Course contents

The course is divided into 4 chapters:
1 - Networks
+ General concepts, ISO Architecture, TCP/IP, Ethernet
2 - Introduction to Operating Systems
+ Main functions of the Operating System, Architecture
+ Commands in Terminal
3 - Cloud
+ General concepts
+ Main architectures
4 - Quantic Architectures
+ General concepts

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	22 hrs	8 hrs	0 hrs	0 hrs	2 hrs

Information Systems [SYSIN]

LEAD PROFESSOR(S): Jean-Yves MARTIN

Requirements

Objectives

The aim of this course is to understand information systems, how they are built, how they can be analyzed.

Course contents

This course focus on following points :

- Structure and management of an Information System
- Cartography of an Information System, Urbanization of Information Systems
- Managing Information Systems Projects
- Data management.
- Risk management

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	20 hrs	10 hrs	0 hrs	0 hrs	2 hrs

Language Theory [TLANG]

LEAD PROFESSOR(S): Didier LIME

Requirements

Objectives

The objective of this course is to introduce a number of fundamental theoretical models of Computer Science, through the notions of languages and compilation.

Course contents

The course follows the classical approach to compilation, extending to the Turing machine models.

1. Lexical analysis, regular expressions, and finite automata
2. Syntax analysis, formal grammars, and pushdown automata
3. The link between languages and algorithms, and Turing machines
4. Semantic analysis and attributed grammars
5. Code generation

These notions are illustrated and put into action through the use of Lex & Yacc derived compiler compilers.

Course material

Alfred V. Aho, Monica S. Lam, Ravi Sethi. Compilers: Principles, Techniques, and Tools (2nd edition). Addison Wesley. 2006

P. Dehornoy. Complexité et Décidabilité, Springer-Verlag, 1993.

M. Sipser. Introduction to the Theory of Computation, PWS Pub. Co., 1996.

Assessment

Individual assessment: EVI 1 (coefficient 1.0)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	3	20 hrs	10 hrs	0 hrs	0 hrs	2 hrs

ENGINEERING PROGRAMME

2024-2025

Year 3

Professional Option

Proficiency in project managament

OP PERFECT

PROGRAMME SUPERVISOR
Thomas LECHEVALLIER

Autumn Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 92	4	Core course	CFOND PROJAV	Core skills consolidation Advanced project management

Spring Semester

Course unit	ECTS Credits	Track	Course code	Title
UE 102	1	Core course	PRODIR PROJPERF	Programme and project management (PMO) Leading a project

Core skills consolidation [CFOND]

LEAD PROFESSOR(S): Thomas LECHEVALLIER

Requirements

- Project management concepts
- development of a GANTT

Objectives

At the end of the course, participants will be able to :
- Understand project governance and roles,
- Plan a project, assess and allocate workloads, and monitor progress,
- Manage risks,
- Analyze the existing situation, express a need, define a target and share a trajectory,

Course contents

- Project definition and project management,
- Methodology for moving from idea to project,
- Reminder of typical project phases and associated deliverables,
- Understanding project governance, scope, schedule and risks,
- Estimating costs and drawing up a budget,
- Project management methods (estimating durations, deadlines, workloads, progress monitoring, work assignment, risk monitoring, expenditure and return on investment monitoring, etc.),
- Reminder of Quality in projects

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	2	22 hrs	8 hrs	0 hrs	2 hrs	0 hrs

Advanced project management [PROJAV]

LEAD PROFESSOR(S): Thomas LECHEVALLIER

Requirements

Objectives

- Sharing a vision,
- Driving change,
- Building a master plan,

Course contents

- Understand and define corporate strategy,
- Distinguish between projects, programs and portfolios,
- Build a Master Plan:
o map and schedule the projects to be managed,
o identify project players: contributions, roles and missions, and multi-project assignments,
o understand the company's culture,
o share vision and work methods,
- Managing a project or program (vision, embodiment, styles and representation),
- Distinguish between and combine organizational and project management,
- Reconciling operational maintenance and project activities,
- Mastering the power of doing, increasing influence and impact,
- Corporate communication,
- Risk management,

Course material

Assessment

Collective assessment: EVC 1 (coefficient 1)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	2	22 hrs	8 hrs	0 hrs	2 hrs	0 hrs

Programme and project management (PMO) [PRODIR]

LEAD PROFESSOR(S): Thomas LECHEVALLIER

Requirements

Objectives

- Finance a project or program and monitor the budget,
- Manage people and communicate throughout the company,
- Leading and motivating project teams,
- Correct unprofessional behavior, manage conflicts,

Course contents

- Management (from self to others)
o Organizing (recruiting, organizing and managing a project team),
o Deciding (gathering information, hypotheses, cognitive biases, sharing and carrying the decision)
o Leading and communicating (leading a team, hierarchy and project management, convincing, presenting, managing conflicts and unprofessional behavior),
o Control (setting up indicators, measuring performance, individual follow-up),
o Anticipate (vision, training and change management),
- Customer relations (analysis of existing situation, expression of needs, definition of target and sharing of trajectory),
- Contractualization (purchasing, contractual procedures, selection of partners, service providers and subcontractors, contract negotiation),
- Budgeting (financial surveys, justification of expenditure, arbitration, budget safeguards).
- Enhancing the value of a project, passing on to operating teams, and closing a project

Course material

Assessment

Individual assessment: EVI 1 (coefficient 1)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	0.5	22 hrs	8 hrs	0 hrs	2 hrs	0 hrs

CENTRALE NANTES

ENGINEERING - OP PERFECT

Year 3 - Spring Semester - Course Unit 102

Leading a project [PROJPERF]

LEAD PROFESSOR(S): Thomas LECHEVALLIER

Requirements

Objectives

Carry out a project in a real-life situation with an identified professional customer

Course contents

Carry out the project throughout the year and apply each notion worked on to the current project.

Course material

Assessment

Collective assessment: EVC 1 (coefficient 1)

LANGUAGE OF INSTRUCTION	ECTS CREDITS	LECTURES	TUTORIALS	LAB	PROJECT	EXAM
French	0.5	0 hrs	0 hrs	0 hrs	34 hrs	0 hrs